

23972 - Digital Data Transmission

Teaching Guide Information

Academic Course: 2026/27

Academic Center: 337 - Engineering School

Study: 3379 - Bachelor's degree in Telecommunications Network Engineering

Subject: 23972 - Digital Data Transmission

Credits: 5.0

Course: 3

Teaching languages:

Theory: Group 1: English

Practice: Group 101: English

Group 102: English

Group 103: English

Group 104: Pending

Seminar: Group 101: English

Group 102: English

Group 103: English

Group 104: Pending

Teachers: Josep Font Segura

Teaching Period: First term

Presentation

Digital data transmission studies the mathematical foundations of reliably sending information over a communication channel. Building on signal theory and information theory, it formalises how data are represented by signals, how they are optimally detected in the presence of noise, and what the limits of reliable transmission are. This discipline underpins practically all of today's communications: mobile telephony (4G and 5G), Wi-Fi, satellite communication, digital television (DVB) and optical fibre. The aims of the course are to become familiar with the mathematical models of signals and channels, to understand optimal detection and digital modulations, and to learn the fundamental limits of transmission (channel capacity).

Associated skills | General learning outcomes

Transverse skills

Instrumental

1. Abstraction capacity to understand and analyze communication systems.
2. Ability to express ideas and concepts accurately orally and in writing.
3. Ability to translate ideas and concepts into a mathematical model and to describe the solutions, either with a rigorous demonstration or with a computer simulation.

Interpersonal

4. Ability to work as a team to solve problems and deepen theoretical contents.
5. Ability to communicate ideas accurately, both orally and in writing.

Systemic

6. Ability to work autonomously to solve a problem.
7. Ability to work systematically, checking and interpreting solutions.

Specific skills

1. Know and apply the basic concepts of communication theory covered in the course.
2. Solve the problems related to communication theory presented in the course.

3. Identify and justify the application of the appropriate mathematical model to analyze a problem and find the solution of it.

In terms of specific skills for the degree, this subject develops CE7, "abstractly model the different elements of a telecommunication system using information theory concepts and methods."

Learning outcomes | Specific learning outcomes

At the end of the course, and regarding the course contents, the student shall be able to:

1. Describe several channel codes for the Gaussian channel and relate their performance, both with formulas and with simulations, with the concept of mutual information.

In addition to this basic knowledge, at the end of the course, the student should be able to:

1. Express with correctness, precision and rigor mathematical ideas and concepts;

2. Use the appropriate software to describe and study the techniques of the course.

The learning outcomes associated to the specific skill CE7 are the following:

RA.CE7.1 Analyze and design simple information compression and coding systems.

RA.CE7.2 Analyze and design communication systems applying information theory.

Contents

The course covers the mathematical foundations and the development and implementation of digital data transmission techniques: the modelling of signals and the Gaussian channel (signal space, energy and noise model), optimal detection in the presence of noise (such as maximum-likelihood or MAP criteria, decision regions and error probability), mutual information and channel capacity, and modulation and channel coding and their performance (for example, modulations such as PAM, PSK or QAM, the matched filter, the Nyquist criterion and bandwidth, multicarrier schemes such as OFDM, and modern error-correcting codes such as polar codes, as well as multi-antenna techniques such as MIMO). These concepts can be linked to real systems such as Wi-Fi, DVB or 4G/5G.

Evaluation and grading system

The course offers two assessment modalities, Modality A and Modality B. Modality B applies by default; a student who prefers Modality A must notify the teaching staff before the end of week 2 of the course. The chosen modality applies for the whole term.

Modality A. It consists of two exams. The midterm exam (P) is worth 20% of the final mark: a multiple-choice test on theory, with problems drawn from a collection of 25 theory problems published at the start of the course on Aula Global; it is held in November and is not re-sittable. The final exam (E) is worth 80% of the final mark: problems drawn from a collection of 50 mathematical problems published at the start of the course on Aula Global, and it is re-sittable. The final mark is $NF = 0.625 \cdot (0.20 \cdot P + 0.80 \cdot E)$; the maximum mark in this modality is 6.25 and the course is passed if $NF \geq 5$.

Modality B. It consists of the following activities. The final exam (E) is worth 60% of the final mark and contains questions on all the contents covered in the course; it is re-sittable. The submissions (D) are worth 30% of the final mark and are transversal submissions in lectures, practicals, seminars and outside the classroom, both in groups and individual. The tests (T) are worth 10% of the final mark and may be real-time gamified questions, Aula Global quizzes or paper-based quizzes, answered individually. To pass this modality, the final exam (E) must be 5 or higher; if $E < 5$, the course's final mark is that of the final exam ($NF = E$). If $E \geq 5$, the final mark is $NF = 0.60 \cdot E + 0.30 \cdot D + 0.10 \cdot T$.

In all assessable activities, plagiarism and academic-integrity breaches are handled according to School and University rules.